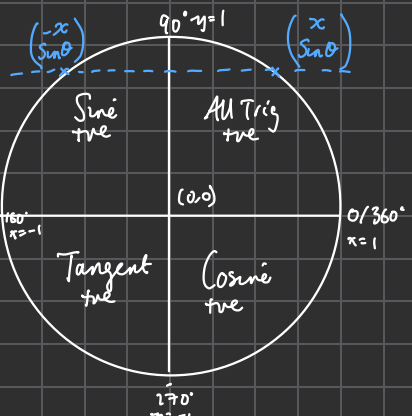


6.1: Trigonometric Functions:

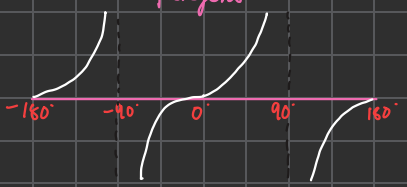
$\sin\theta = \frac{O}{H}$ $\cos\theta = \frac{A}{H}$ $\tan\theta = \frac{O}{A}$

$0° - 0$	$0° - 1$	$0° = 0$
$30° - \frac{1}{2}$	$30° - \frac{\sqrt{3}}{2}$	$30° - \frac{1}{\sqrt{3}}$
$45° - \frac{\sqrt{2}}{2}$	$45° - \frac{\sqrt{2}}{2}$	$45° - 1$
$60° - \frac{\sqrt{3}}{2}$	$60° - \frac{1}{2}$	$60° - \sqrt{3}$
$90° - 1$	$90° - 0$	$90° -$ undefined









In this diagram, $\cos(\theta)$ is seen as the x-coordinate, and $\sin(\theta)$ seen as y, hence points are shown as $(\cos\theta, \sin\theta)$

This is why ASTC is applied. This also shows why for each value of a trig function, there are 2 solutions (from $0° \le \theta \le 360°$)

To figure out values in the unit circle, find an initial angle for the trig value, then find another one based on ASTC & the fact of reference angles:

QI - θ, QII - $180 - \theta$, QIII - $180 + \theta$, QIV - $360 - \theta$

To find angles from graphs:

- Draw graph
- Sketch a line for all desired trig values. Find where they are on the x-axis.

Sine Features: $P = 360°$

$\sin(\theta) > 0 \Rightarrow \sin(\theta) = \sin(180 - \theta)$

$\sin(\theta) < 0 \Rightarrow \sin(\theta) = \sin(540 - \theta)$

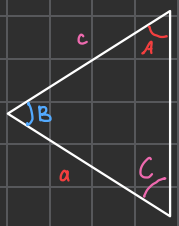
Cosine Features: $P = 360°$

$\cos\theta > 0 \Rightarrow \cos\theta = \cos(360 - \theta)$

$\cos\theta < 0 \Rightarrow \cos\theta = \cos(360 - \theta)$

Tangent Features: $P = 180°$

Triangles without Right-Angles:



Sine Rule

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

This produces acute angles. To obtain other types of angles, perform the obvious operation for positive sine values.

Cosine Rule

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$A = \cos^{-1}\left(\frac{b^2 + c^2 - a^2}{2bc}\right)$$

Area Rule

$$A_{\text{triangle}} = \frac{1}{2}bc\sin A$$